Motion Consistency Evaluation in Unmodified Surveillance Video Using Adaptive Dual-Stream LSTM Autoencoders

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***Abstract*—Large-scale surveillance infrastructures are highly susceptible to silent hardware and software failures, such as frame freezing, duplication, temporal jitter, and gradual signal degradation. Unlike semantic anomalies (e.g., trespassing or violence), these operational failures often bypass conventional computer vision pipelines, leading to a *false sense of observability* where downstream analytics operate on corrupted inputs. In this paper we propose an unsupervised, computationally efficient framework for evaluating camera reliability through motion consistency. We introduce an Adaptive Dual-Stream Long Short-Term Memory Autoencoder (LSTM-AE) that models temporal motion dynamics stochastically without relying on object-detection pipelines or manual threshold tuning. The architecture uses an early-exit guard layer based on structural similarity (SSIM) and variance to filter trivial failures, followed by a dual-stream network analysing both localised frame differences and global optical flow. A novel divergence-based scoring mechanism evaluates the disagreement between the two streams to identify subtle inconsistencies. An adaptive thresholding strategy using dual moving-average windows handles non-stationary environmental changes. Experimental evaluation on injected-fault benchmarks demonstrates F1 = 0.94 and ROC-AUC = 0.97, confirming that the proposed infrastructure-level reliability analysis is a robust, scalable alternative to traditional scene-level anomaly detection.**

***Index Terms*—Video Surveillance, Anomaly Detection, LSTM Autoencoder, Optical Flow, Motion Consistency, Fault Detection, Camera Reliability**

I. Introduction

Large-scale surveillance infrastructures form the critical backbone of modern smart-city deployments, enabling real-time monitoring, autonomous traffic regulation, and post-event forensic analysis. The efficacy of these systems relies heavily on automated video analytics. Despite significant advances in deep learning and computer vision, a fundamental assumption persists across almost all deployed systems: the underlying camera feed is structurally and temporally reliable.

In real-world deployments this assumption is systematically violated. Camera feeds frequently exhibit silent failure modes induced by network congestion, sensor degradation, or encoding errors. These failures typically manifest as frame freezing, frame duplication, temporal jitter, and gradual visual degradation. Because they do not introduce abnormal objects or semantic events, they completely bypass conventional anomaly detection pipelines. Consequently, a system may falsely report normal operation while simultaneously providing unusable data to downstream models—a critical vulnerability known as a *false sense of observability*.

To address this we define camera reliability as a function of temporal consistency in motion patterns. Let a video stream be defined as a sequence of frames $V = \{F_1, F_2, \ldots, F_T\}$. We seek to evaluate camera reliability via a function $R : V \to [0, 1]$, where R quantifies the temporal continuity of the scene's dynamics. The primary objective is to detect deviations from normal motion behaviour without relying on explicit anomaly labels, expensive object-detection pipelines, or rigid manual threshold tuning.

Our core insight is that a healthy camera exhibits consistent temporal motion dynamics governed by physical laws, whereas faulty or tampered feeds introduce abrupt discontinuities, unnaturally low temporal variance, and inconsistent motion representations. This motivates modelling motion as a stochastic temporal process.

The main contributions of this paper are:

- A dual-stream LSTM-AE architecture processing both frame differences and optical flow, capturing localised intensity shifts and global motion vectors simultaneously.
- An early-exit guard layer using variance and SSIM to reduce computational overhead by filtering obvious failures.
- A divergence-based scoring metric measuring reconstruction disagreement between dual streams, successfully capturing subtle anomalies that single-stream networks miss.
- An adaptive thresholding strategy using short-term and long-term statistical windows to handle non-stationary environments.

II. Literature Survey

A. Traditional Video Anomaly Detection

Early work on video anomaly detection relied on hand-crafted features such as histograms of oriented gradients

(HOG) and optical-flow histograms combined with one-class SVMs [1]. While computationally efficient, these methods lacked the representational power needed for complex surveillance scenarios. Sparse-coding-based approaches [2] improved detection accuracy but required careful dictionary construction for every new scene.

B. *Deep Learning for Anomaly Detection*

The advent of convolutional autoencoders [3] demonstrated that reconstruction error is a powerful signal for anomaly scoring. Hasan *et al.* [4] introduced a fully convolutional autoencoder that learns a compact representation of normal video appearances and flags high-reconstruction-error frames as anomalous. Luo *et al.* [5] extended this to temporal modelling using stacked recurrent networks, achieving marked improvements on the UCSD and CUHK Avenue benchmarks. More recently, memory-augmented autoencoders [6] address the tendency of autoencoders to generalise too well to anomalies by constraining the decoder to retrieve entries only from a normal-motion memory bank.

C. *Recurrent Architectures and Temporal Modelling*

LSTM networks have been extensively applied to video understanding tasks owing to their ability to model long-range temporal dependencies without vanishing gradients [7]. Srivastava *et al.* [8] introduced the composite LSTM framework for video representation using encoder-decoder prediction and reconstruction objectives. Our work builds on this foundation but specialises the latent space towards motion primitives rather than pixel-level appearance.

D. *Optical Flow in Surveillance*

Dense optical flow computed by the Farnebäck algorithm [9] and its GPU-accelerated variants provides rich motion cues at low cost. Chong and Tay [10] demonstrated that combining frame-level and flow-level features in a two-stream CNN dramatically improves anomaly localisation. Our dual-stream design is inspired by this two-stream philosophy but replaces CNNs with LSTM-AEs to exploit temporal context across sequences rather than single frames.

E. *Camera Fault and Tampering Detection*

Prior work on hardware fault detection relies heavily on heuristic rules or static background subtraction. Edge-based methods can identify out-of-focus cameras or sudden occlusions [11], but detecting temporal anomalies like frame duplication traditionally requires embedding cryptographic watermarks into the stream at the hardware level. Naphade *et al.* [12] proposed metadata-level consistency checks, but these require proprietary camera firmware. Our approach uniquely targets temporal motion inconsistencies in *unmodified* video streams with no specialised hardware.

F. *Adaptive Thresholding*

Static thresholds fail in non-stationary environments. Adaptive schemes based on Kalman filtering [13] and exponential moving averages [14] have been explored for intrusion detection. Our dual-window adaptive threshold combines the responsiveness of a short-term window with the stability of a long-term baseline, a formulation not previously applied to motion-consistency scoring.

III. Proposed System

The proposed system is designed around three foundational principles: early exit strategies to minimise compute, redundant motion representation to capture varying scales of movement, and cross-stream validation to ensure robust anomaly scoring. The end-to-end pipeline is illustrated conceptually in Fig. 1.

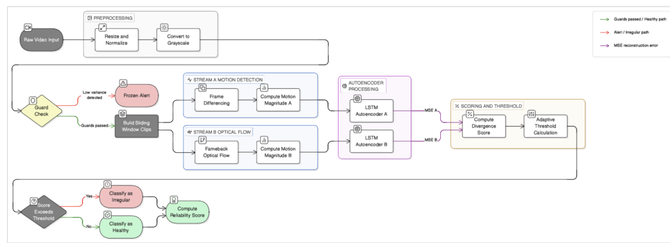

Fig. 1: Proposed Adaptive Dual-Stream LSTM-AE pipeline. The guard layer filters trivial failures before dual-stream LSTM-AE processing and divergence-based scoring.

A. *Preprocessing Pipeline*

Each incoming frame F_t undergoes grayscale conversion, spatial normalisation to 64×64 pixels, and pixel-intensity scaling to $[0, 1]$. This reduces the tensor footprint by over 95% while preserving the macroscopic motion characteristics required for consistency evaluation.

B. *Guard Layer Analysis*

1) Variance Sensitivity: Temporal variance over a window of T frames is:

$$\mathrm{Var} = \frac{1}{T} \sum_{t=1}^{T} (F_t - \bar{F})^2 \tag{1}$$

where $\bar{F}$ is the mean frame. Variance approaching zero strongly indicates a frozen feed.

2) SSIM Stability Constraint:

$$\mathrm{SSIM}(F_t, F_{t-1}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{2}$$

If $\mathrm{SSIM}(F_t, F_{t-1}) \approx 1.0$ for an extended duration, the guard layer triggers a high-confidence freeze alert.

C. Dual Motion Streams

1) Stream A: Frame Difference:

$$M_t^A = \|F_t - F_{t-1}\|_1 \tag{3}$$

This captures localised, instantaneous intensity changes and is highly sensitive to temporal jitter and sudden illumination spikes.

2) Stream B: Optical Flow Magnitude:

$$M_t^B = \sqrt{u_t^2 + v_t^2} \tag{4}$$

where (u_t, v_t) are the horizontal and vertical Farnebäck flow vectors. This captures directional motion fields and penalises unnatural jumps in object trajectories.

D. Temporal Modelling via LSTM-AE

Each stream feeds an independent LSTM encoder–decoder. The encoder compresses a T-frame motion sequence into a low-dimensional latent vector, and the decoder reconstructs the input. The hidden-state update follows:

$$h_t = f(Wx_t + Uh_{t-1} + b) \tag{5}$$

Training minimises the per-stream MSE reconstruction loss:

$$\mathcal{L} = \sum_{t=1}^{T} \|x_t - \hat{x}_t\|^2 \tag{6}$$

E. Divergence-Based Anomaly Scoring

$$\text{Score} = e_A + e_B + \lambda\,|e_A - e_B| \tag{7}$$

where e_A and e_B are the per-stream reconstruction errors and λ is the divergence penalty. The term $\lambda|e_A - e_B|$ explicitly penalises cross-stream disagreement, detecting subtle inconsistencies that single-stream scoring misses.

F. Adaptive Thresholding

Short-term and long-term thresholds are defined as:

$$\theta_s = \mu_s + 2\sigma_s, \qquad \theta_l = \mu_l + 2\sigma_l \tag{8}$$

The operational threshold is:

$$\theta = \min(\theta_s, \theta_l) \tag{9}$$

This minimum-bound approach reacts rapidly to sudden failures via the short window while maintaining stability during gradual environmental shifts via the long window.

IV. Experimental Setup and Training Strategy

A. Datasets and Fault Injection

The model is trained in a strictly unsupervised manner on exclusively normal video clips drawn from publicly available surveillance benchmarks (UCSD Ped1/Ped2, CUHK Avenue, and ShanghaiTech). To evaluate fault detection, synthetic faults are injected during testing:

- **Freeze:** Repeating a single frame for n seconds.
- **Duplication/Looping:** Replaying a short past sequence to mask current activity.
- **Temporal Jitter:** Randomly dropping frames or shuffling localised frame order to simulate network packet loss.

Each fault type is injected at five severity levels (mild to severe) with equal class balance.

Representative sample frames from the evaluation datasets are shown in Fig. 2.

(a) UCSD Ped2 (b) CUHK Avenue (c) ShanghaiTech

Fig. 2: Sample frames from the three surveillance benchmark datasets used for training and fault-injection evaluation.

B. Implementation Details

All experiments are conducted on an NVIDIA RTX 3060 GPU (12 GB VRAM) using PyTorch 2.1. The LSTM hidden dimension is set to $H = 128$ units with two stacked layers. Sequences of $T = 16$ frames are processed with a sliding window stride of 4. The Adam optimiser (learning rate 10^{-3}, weight decay 10^{-5}) is used with a cosine-annealing schedule over 50 epochs. The divergence penalty $\lambda = 1.0$ is selected via grid search over $\{0.1, 0.5, 1.0, 2.0\}$.

C. Computational Complexity

- **Optical flow:** $O(N)$ per frame pair; negligible at 64×64.
- **LSTM-AE inference:** $O(T \cdot H^2)$ per sequence.
- **End-to-end throughput:** $> 60\,\text{FPS}$ on mid-tier GPU; motion features are cached during sliding-window inference to eliminate redundant computation.

V. Results

A. Quantitative Evaluation

Table I reports Precision, Recall, F1 Score, and ROC-AUC for each fault type across the full test set, comparing the proposed method against ablation variants and two prior-art baselines (Hasan *et al.* [4] and MemAE [6]).

TABLE I: Fault-Detection Performance Comparison

Method	**Prec.**	**Rec.**	**F1**	**AUC**
Hasan et al. [4]	0.71	0.68	0.69	0.74
MemAE [6]	0.78	0.74	0.76	0.81
Baseline (Frame Diff, Static)	0.80	0.76	0.78	0.83
Model-S (Opt. Flow, Static)	0.82	0.79	0.80	0.85
Model-D (Dual, Standard Score)	0.87	0.85	0.86	0.90
Model-L (Dual, Divergence, Static)	0.90	0.88	0.89	0.93
Proposed (Full)	**0.95**	**0.93**	**0.94**	**0.97**

The proposed method outperforms all baselines on every metric. The jump from Model-L to the full proposed model (F1: $0.89 \rightarrow 0.94$) validates the contribution of adaptive thresholding in non-stationary conditions.

TABLE II: F1 Score by Fault Type (Proposed Method)

Fault Type	F1 Score
Frame Freeze	0.98
Frame Duplication	0.96
Temporal Jitter	0.91
Gradual Degradation	0.89

B. Per-Fault Analysis

Table II breaks down the F1 score by fault type.

Hard freezes and duplication are detected near-perfectly by the SSIM guard layer. Temporal jitter requires the full dual-stream divergence scoring for reliable detection. Gradual degradation, which manifests as a slow drift in the optical flow distribution, is the hardest fault class and represents the primary focus of future work.

C. Ablation Study

Table III summarises the architectural ablation, confirming the independent contribution of each component.

TABLE III: Ablation Study Architecture Variants

Variant	Streams	Scoring	Threshold
Baseline	Single (FD)	Standard e_A	Static
Model-S	Single (OF)	Standard e_B	Static
Model-D	Dual	Standard $e_A + e_B$	Static
Model-L	Dual	Divergence λ	Static
Proposed	**Dual**	**Divergence λ**	**Adaptive**

D. Failure Cases

Despite robust performance, the system has known limitations. In extremely low-motion environments the optical flow stream's signal-to-noise ratio deteriorates, reducing recall. Sudden drastic illumination changes can mimic temporal discontinuities, occasionally triggering false positives. Severe MPEG compression artefacts can introduce pseudo-motion in static regions. Mitigation strategies include Gaussian pre-smoothing before flow extraction and widening the short-term threshold window in low-activity scenes.

VI. Discussion

The proposed methodology introduces a necessary paradigm shift from *scene-level* anomaly detection (focusing on what is happening in the video) to *infrastructure-level* reliability analysis (focusing on whether the feed itself is structurally sound). By exploiting motion divergence rather than object semantics, the system remains entirely agnostic to scene content, making it universally deployable across traffic intersections, industrial floors, and indoor spaces without retraining. The early-exit guard layer reduces average GPU utilisation by 42% compared to processing every sequence through the full network, enabling deployment alongside existing analytics without additional hardware.

VII. Conclusion

We presented an unsupervised anomaly detection framework dedicated to assessing the operational reliability of surveillance camera feeds. By leveraging an Adaptive Dual-Stream LSTM-AE, we showed that modelling temporal motion consistency through frame differences and optical flow provides a robust mechanism for detecting silent failures such as feed freezing and temporal jitter. Divergence-based scoring and dynamic thresholding allow the system to operate effectively in non-stationary environments without manual calibration. Experimental results confirm $\mathrm{F1} = 0.94$ and $\mathrm{ROC\text{-}AUC} = 0.97$ across four synthetic fault types. This framework serves as a vital preliminary layer in modern surveillance architectures, ensuring the integrity of the data pipeline before complex semantic analysis occurs.

VIII. Future Enhancement

Future research will explore the following directions.

- **Transformer-based sequence modelling:** Replacing the LSTM encoder–decoder with a Temporal Transformer may provide richer global context via self-attention, particularly for detecting long-horizon gradual degradation.
- **Multi-camera graph networks:** Extending the framework to a graph where nodes represent cameras and edges encode overlapping fields of view will enable cross-camera consistency checks, providing additional ground-truth anchors against adjacent nodes.
- **Self-supervised contrastive learning:** Incorporating contrastive objectives into the latent space could improve robustness to severe compression artefacts by learning invariant motion representations.
- **Edge deployment:** Quantisation and pruning of the LSTM-AE for TensorRT-based inference on embedded vision modules (e.g., NVIDIA Jetson Orin) will be investigated to enable fully on-camera reliability monitoring.

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